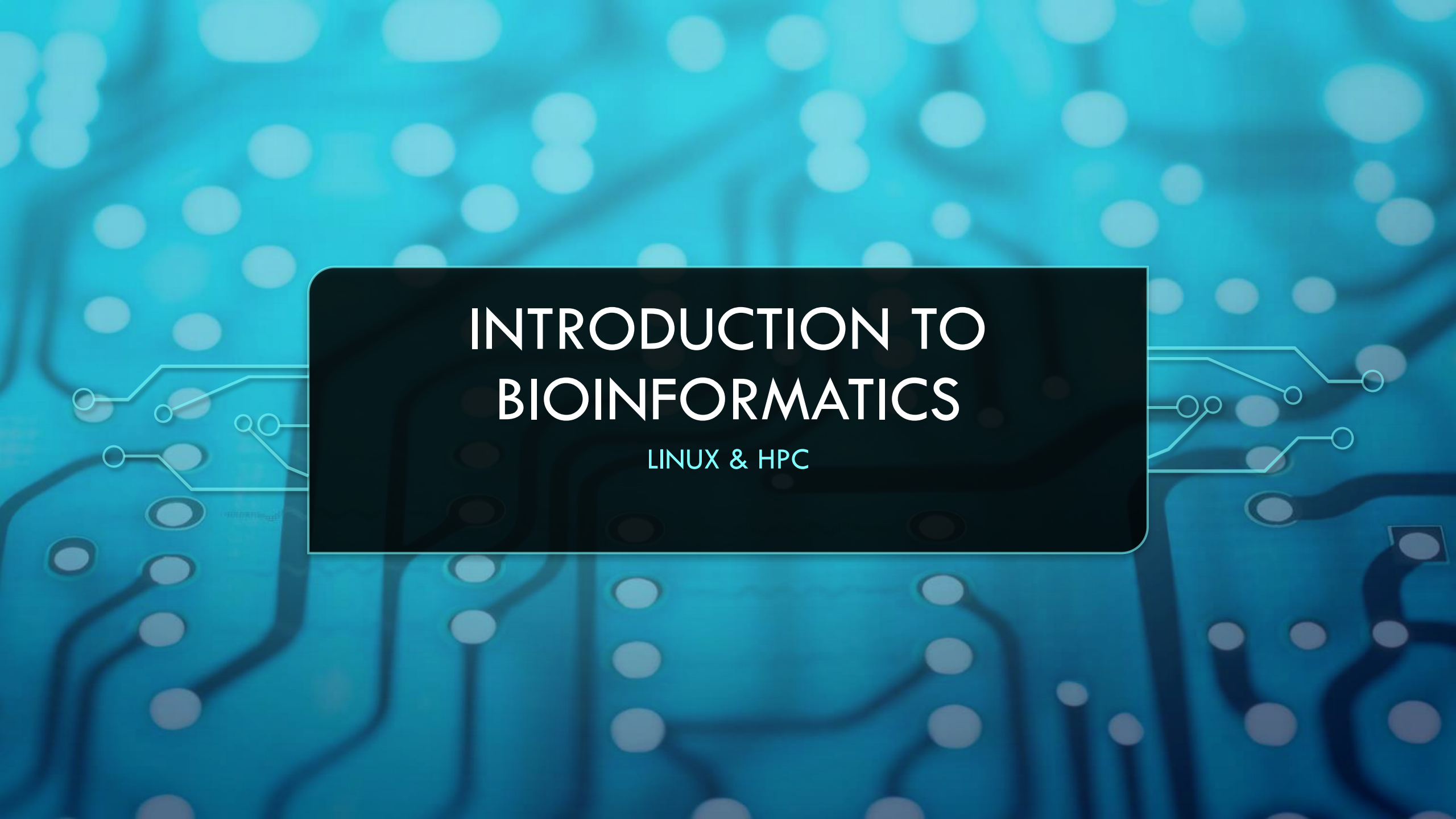




INTRODUCTION TO BIOINFORMATICS

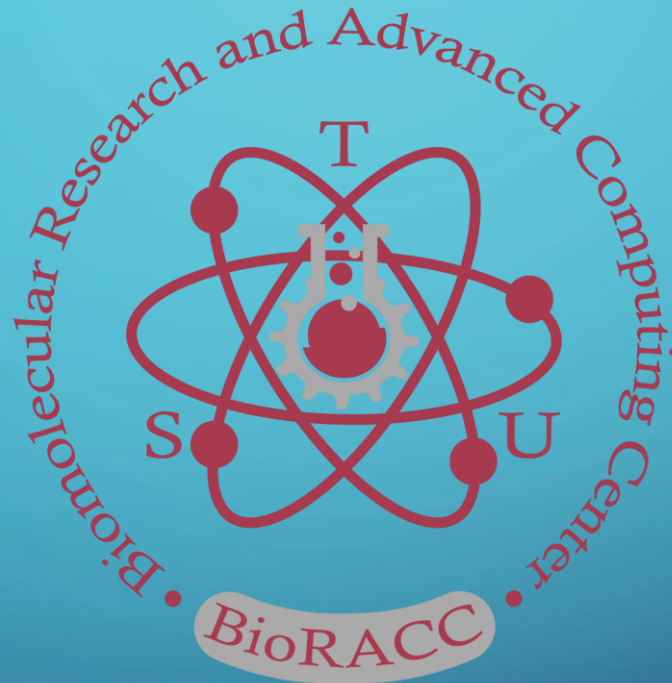
LINUX & HPC

BIOINFORMATICS

- Intersection of biology, statistics/data science, and computer science
- “Omics”:
 - DNA-sequencing – mutations; epigenetics
 - RNA-sequencing – gene expression; isoforms
 - Metabolomics – metabolites/small molecules
 - Proteomics – proteins; PTMs
- Structural biology
- Systems biology



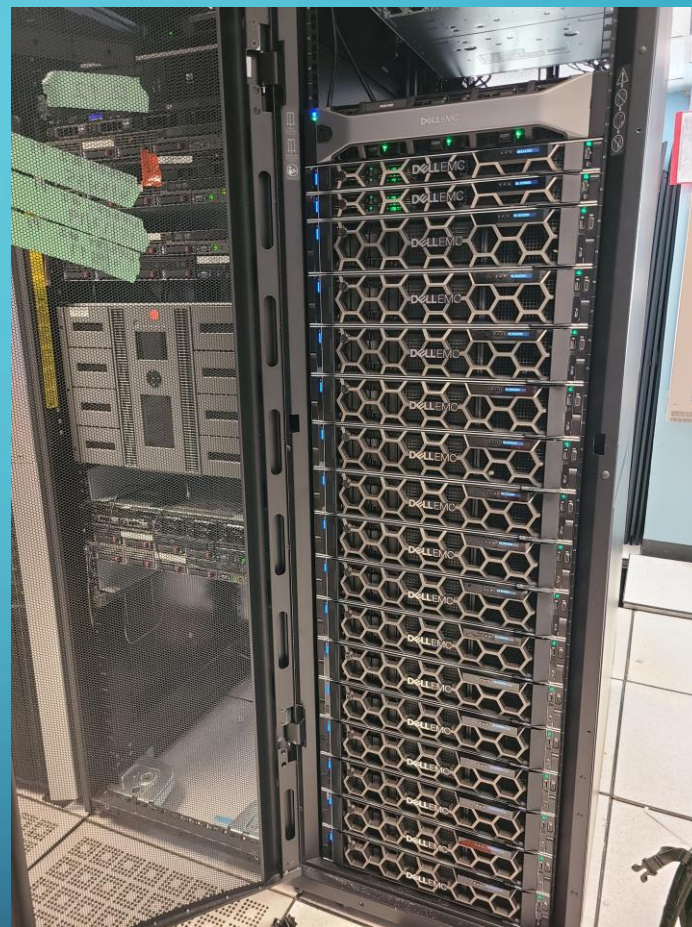
BIORACC HIGH PERFORMANCE COMPUTING CLUSTER (HPC)



Solutions through the Lens of Science

HIGH PERFORMANCE COMPUTING CLUSTER (HPC)

- 15 nodes:
 - 32-core Intel processor
 - NVIDIA A30 Graphics Processing Unit (GPU)
 - 256 GB RAM
- Shared 73 TB memory
- Slurm Workload Manager
- Open OnDemand web portal



SECURE SHELL (SSH)

- Remotely access another computer/server/cluster
- macOS has a built-in SSH client called Terminal
- Windows has PowerShell
- Third party SSH clients have more features (e.g. MobaXterm or PuTTY)
- Public-key cryptography for a secure connection

• Computer 1



• Computer 2



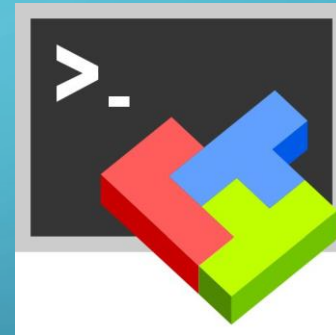
ACCESSING REMOTE SERVERS/CLUSTERS WITH SSH



Mac
Terminal



Windows
PowerShell



MobaXterm



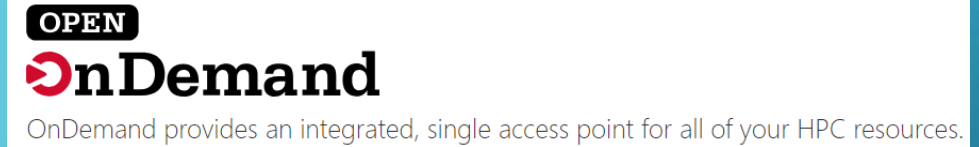
PuTTY

OPEN

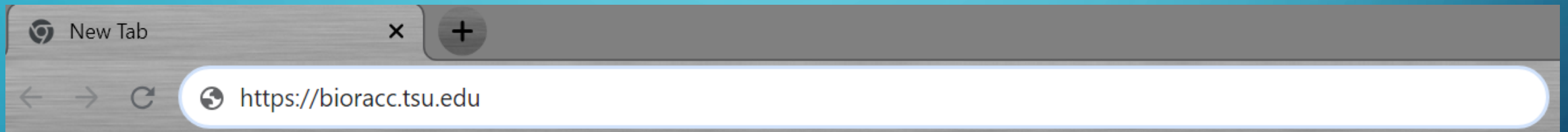
 **nDemand**

OnDemand provides an integrated, single access point for all of your HPC resources.

ACCESSING THE CLUSTER VIA OPEN ONDEMAND



1. Type <https://bioracc.tsu.edu> in a browser



Must be connected to TSU-NET secure network or the VPN to access cluster



LOGGING IN

Username: firstname.lastname (all lowercase)

Default password: firstname.lastname.**changeme**

Sign in to access this site
Authorization required by <https://bioracc.tsu.edu>
Username
Password

SUCCESSFULLY LOGGED IN



TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  



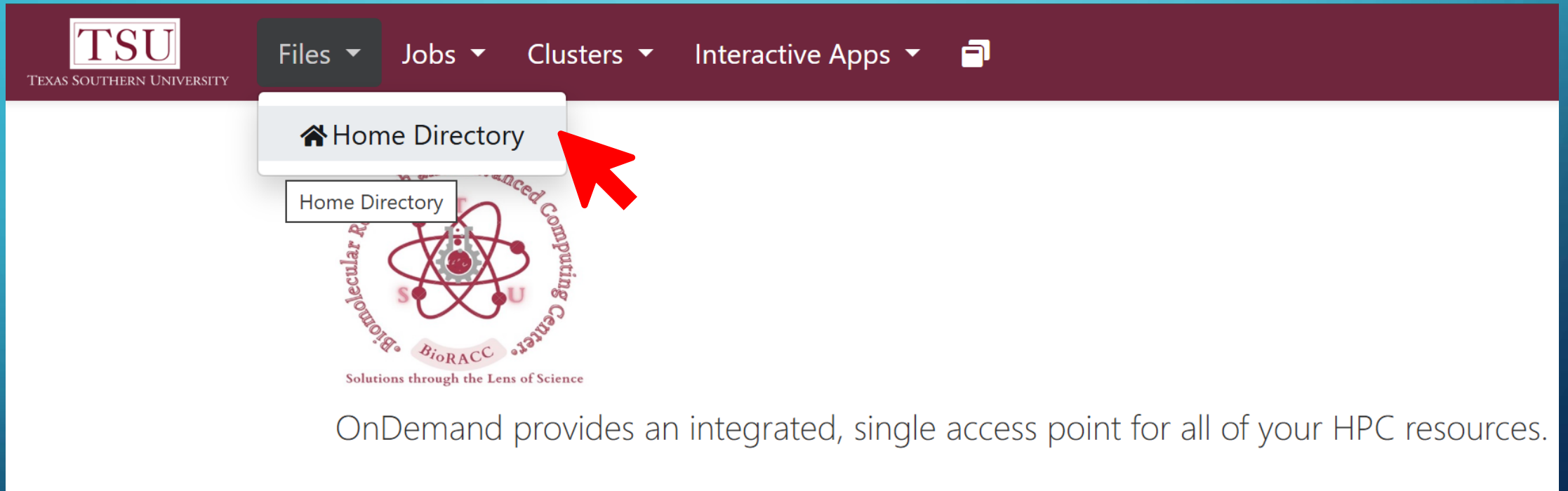
Solutions through the Lens of Science

OnDemand provides an integrated, single access point for all of your HPC resources.

powered by
 

OnDemand version: 3.0.1

HOME DIRECTORY



The screenshot shows the OnDemand web interface. At the top is a dark red navigation bar with the TSU logo (Texas Southern University) on the left and a menu on the right containing 'Files', 'Jobs', 'Clusters', and 'Interactive Apps', each with a dropdown arrow. A red arrow points to the 'Files' dropdown menu, which is open, showing 'Home Directory' as the selected option. Below the navigation bar, the main content area features a large, faint circular logo for the 'BioRACC' (Biomolecular Research Advanced Computing Center) with the text 'Solutions through the Lens of Science'. Below this logo, the text reads: 'OnDemand provides an integrated, single access point for all of your HPC resources.'

TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

Home Directory

Home Directory

BioRACC
Biomolecular Research Advanced Computing Center
Solutions through the Lens of Science

OnDemand provides an integrated, single access point for all of your HPC resources.

CONTENTS OF YOUR HOME DIRECTORY

 TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  

>_ Open in Terminal ▾

 Refresh

 New File

 New Directory

 Upload

 Download

 Copy/Move

 Delete

Home Directory

 / home / databiom /

 Change directory

 Copy path

☐ Show Owner/Mode ☐ Show Dotfiles Filter:

Showing 1 of 11 rows - 0 rows selected

powered by  OnDemand

OnDemand version: 3.0.1

ACTIVE JOBS

TSU

TEXAS SOUTHERN UNIVERSITY

Files ▾

Jobs ▾

Clusters ▾

Interactive Apps ▾

 ▾





⌚ Active Jobs

🔧 Job C...oser

Active Jobs

➤ Open in Terminal

↻ Refresh

+ New File

➤ New Directory

⬆ Upload

⬇ Download

📄 Copy/Move

🗑 Delete

Home Directory

⬆

/ home / databiom /

📄 Change directory

📄 Copy path

☐ Show Owner/Mode

☐ Show Dotfiles

Filter:

Showing 1 of 11 rows - 0 rows selected

	Type	Name	Size	Modified at
☐	📁	ondemand	-	1/26/2025 12:00:22 PM

powered by

OPEN

OnDemand

OnDemand version: 3.0.1

SEE RUNNING OR QUEUED JOBS



Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

Your Jobs ▾All Clusters ▾

Active Jobs

Show

50 ▾

 entries

Filter:

ID	Name	User	Account	Time Used	Queue	Status	Cluster	Actions
No data available in table								

Showing 0 to 0 of 0 entries

PreviousNext

powered by



OnDemand version: 3.0.1

JOB COMPOSER

 **TSU**
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  

Your Jobs ▾ All Clusters ▾

Filter:

Active Jobs

Show

50 ▾

 entries

⌚ Active Jobs

 Job Composer

Job Composer

ID ⌕⁻²
⌕^{-1/4}

Name

User ⌕⁻²
⌕^{-1/4}

Account ⌕⁻²
⌕^{-1/4}

Time Used ⌕⁻²
⌕^{-1/4}

Queue ⌕⁻²
⌕^{-1/4}

Status ⌕⁻²
⌕^{-1/4}

Cluster ⌕⁻²
⌕^{-1/4}

Actions ⌕⁻²
⌕^{-1/4}

No data available in table

Showing 0 to 0 of 0 entries

Previous

Next

powered by  **OnDemand**

OnDemand version: 3.0.1

COMPOSE A “JOB”

Open OnDemand / Job Composer

Jobs

Templates

? Help

Jobs

+ New Job ▾

☆ Create Template

✎ Edit Files

⚙ Job Options

🔗 Open Terminal

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

Created	↓↑ Name	↓↑ ID	↓↑ Cluster	↓↑ Status	↓↑
January 26, 2025 12:15pm	(default) Simple Sequential Job		SLURM Cluster	Not Submitted	

Showing 1 to 1 of 1 entries

Previous 1 Next

Job Details

Job Name:

(default) Simple Sequential Job

Submit to:

SLURM Cluster

Account:

Not specified

Script location:

/home/databiom/ondemand/data/sys/myjobs/projects/default/1

SHELL ACCESS



Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

>_SLURM Cluster Shell Access

SLURM Cluster Shell Access

Your Jobs ▾ All Clusters ▾

Active Jobs

Show

50 ▾

 entries Filter:

ID ^{â-2} _{â-1/4}	Name	User ^{â-2} _{â-1/4}	Account ^{â-2} _{â-1/4}	Time Used ^{â-2} _{â-1/4}	Queue ^{â-2} _{â-1/4}	Status ^{â-2} _{â-1/4}	Cluster ^{â-2} _{â-1/4}	Actions ^{â-2} _{â-1/4}
No data available in table								

Showing 0 to 0 of 0 entries Previous Next

powered by



OnDemand version: 3.0.1

TERMINAL OPENS IN NEW TAB

Host: localhost

databiom@localhost's password:

Last login: Sun Jan 26 11:57:54 2025 from 192.168.222.97

[databiom@tsu-hpc01 ~]\$

CHANGING YOUR PASSWORD

Host: localhost

databiom@localhost's password:

Last login: Sun Jan 26 11:57:54 2025 from 192.168.222.97

[databiom@tsu-hpc01 ~]\$ passwd

Changing password for user databiom.

Current Password:

New password:

Retype new password:

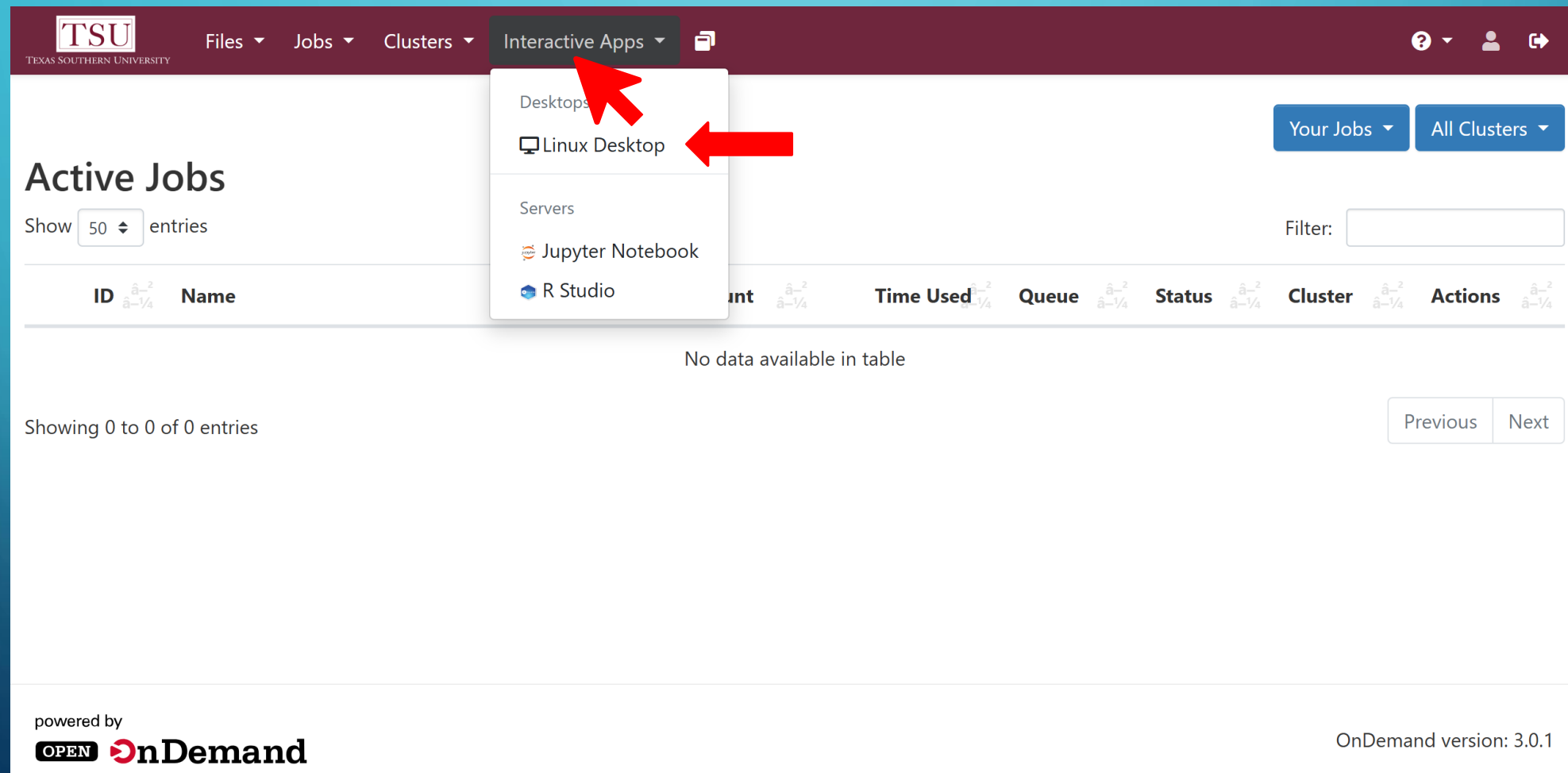
passwd: all authentication tokens updated successfully.

[databiom@tsu-hpc01 ~]\$



Text will not show up

REQUESTING A LINUX DESKTOP SESSION



The screenshot displays the TSU OnDemand web interface. The top navigation bar includes links for Files, Jobs, Clusters, and Interactive Apps. The 'Interactive Apps' dropdown menu is open, showing options: Desktops, Linux Desktop (highlighted with a red arrow), Servers, Jupyter Notebook, and R Studio. The main content area is titled 'Active Jobs' and shows a table with columns: ID, Name, Amount, Time Used, Queue, Status, Cluster, and Actions. The table is currently empty, displaying 'No data available in table'. The footer includes the TSU logo, 'powered by OPEN OnDemand', and 'OnDemand version: 3.0.1'.

TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

Desktops
Linux Desktop
Servers
Jupyter Notebook
R Studio

Your Jobs ▾ All Clusters ▾

Active Jobs

Show 50 entries

Filter:

ID	Name	Amount	Time Used	Queue	Status	Cluster	Actions
No data available in table							

Showing 0 to 0 of 0 entries

Previous Next

powered by
OPEN OnDemand

OnDemand version: 3.0.1

REQUESTING A LINUX DESKTOP SESSION

 **TSU**
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

? ▾  

Home / My Interactive Sessions / Linux Desktop

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop

This app will launch an interactive desktop on a compute node. You will have full access to the resources defined below. This is analogous to an interactive batch job.

How many resources do you need for this desktop?

☒ 4 Cores 

☐ 8 Cores

☐ 16 Cores, 1 GPU

☐ 32 Cores, 1 GPU

How many hours do you need the resources for?



 **Launch** 

* The Linux Desktop session data for this session can be accessed under the [data root directory](#).

REQUESTING A LINUX DESKTOP SESSION

TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

? ▾  

Session was successfully created. 

Home / My Interactive Sessions

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop (4201) Queued

Created at: 2025-01-26 12:33:43 CST  Delete

Time Requested: 2 hours

Session ID: [cf54a055-29d2-411d-9a83-e3fa32633f77](#)

Please be patient as your job currently sits in queue. The wait time depends on the number of cores as well as time requested.

powered by
 

OnDemand version: 3.0.1

LAUNCH LINUX DESKTOP SESSION



TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  

Session was successfully created. 

Home / My Interactive Sessions

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop (4201) 1 node | 4 cores | Running 

Host: >_gpu001.cm.cluster  Delete

Created at: 2025-01-26 12:33:43 CST

Time Remaining: 1 hour and 59 minutes

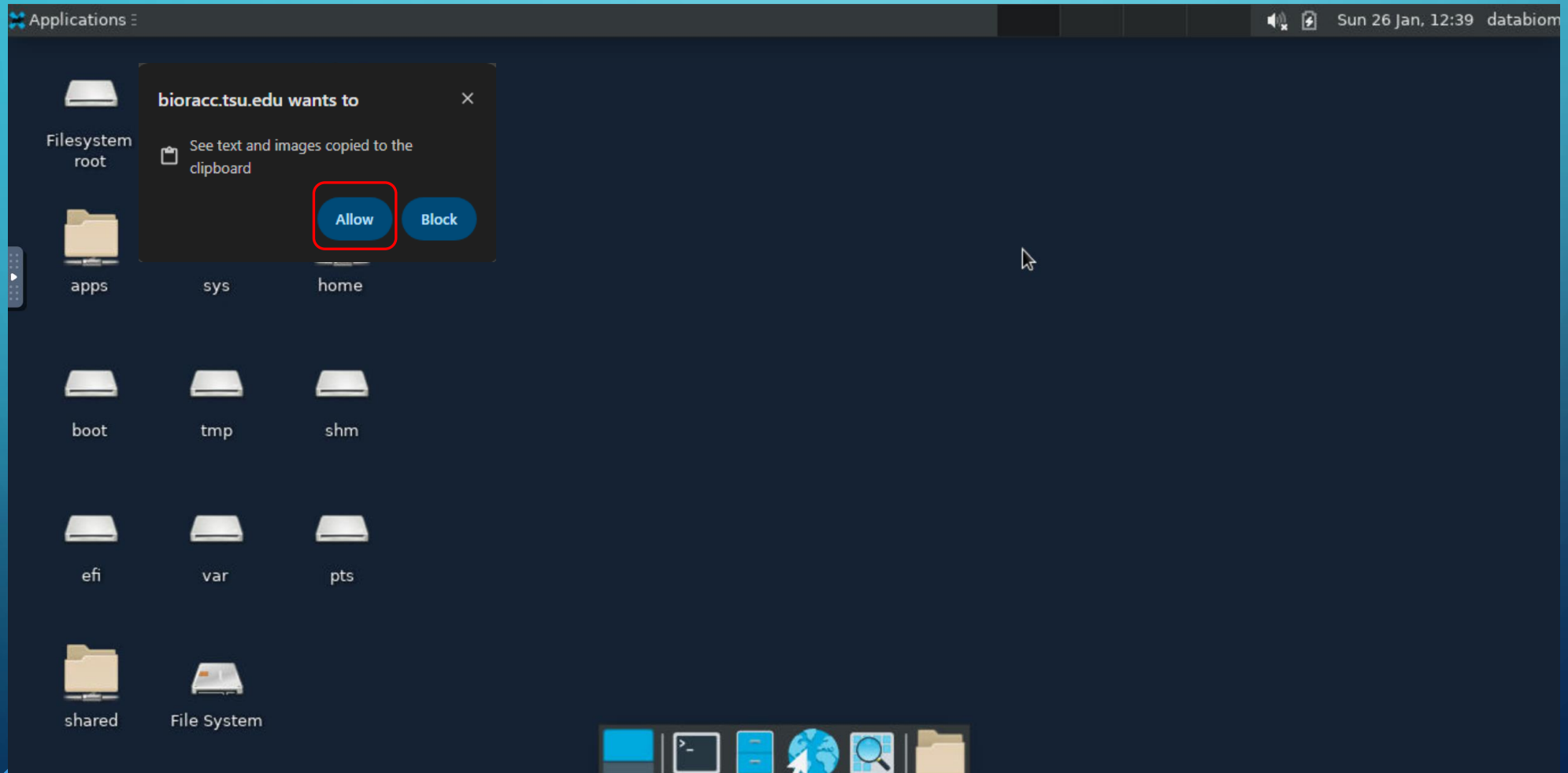
Session ID: cf54a055-29d2-411d-9a83-e3fa32633f77

Compression  0 (low) to 9 (high)

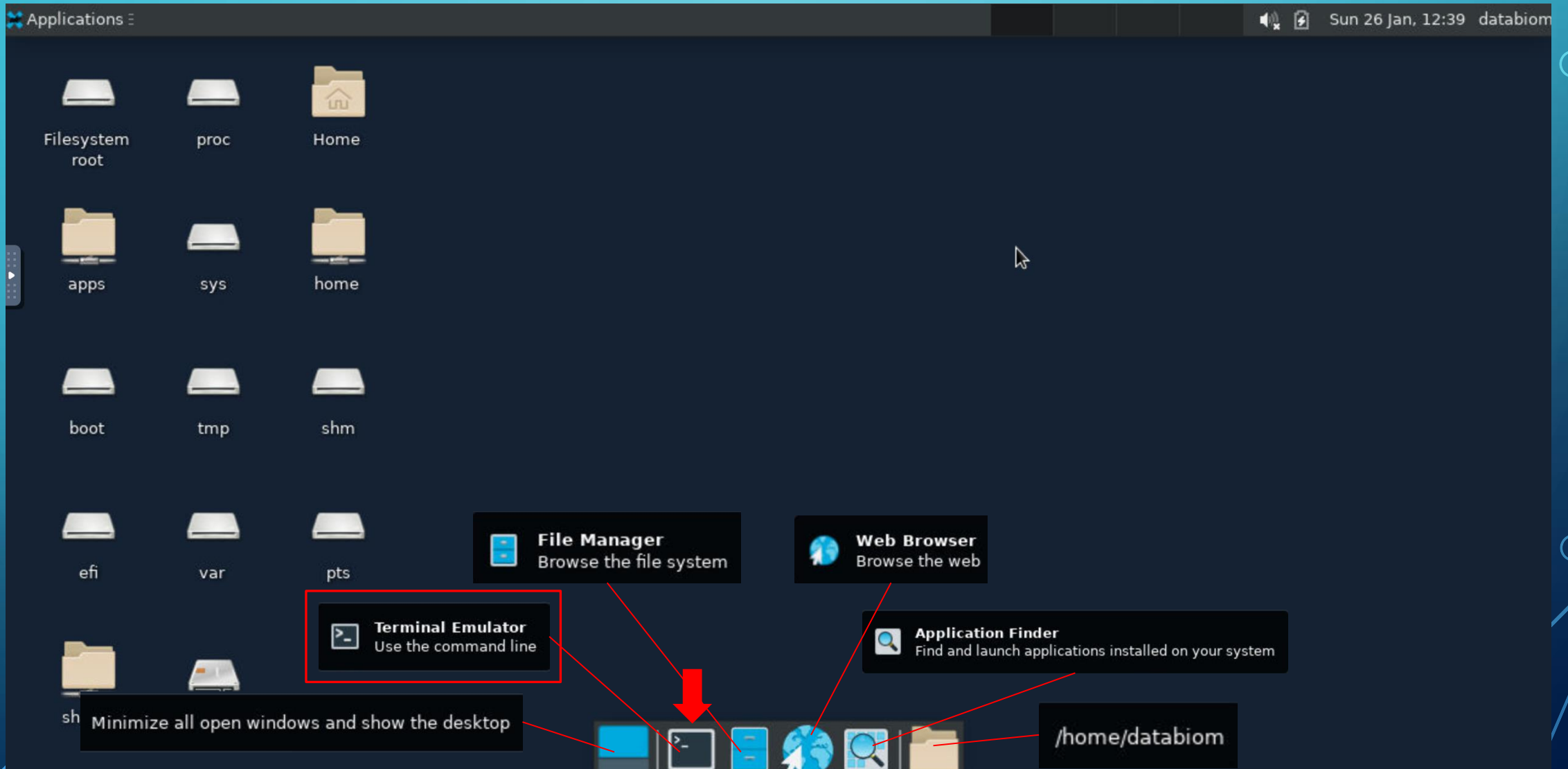
Image Quality  0 (low) to 9 (high)

 Launch Linux Desktop View Only (Share-able Link)

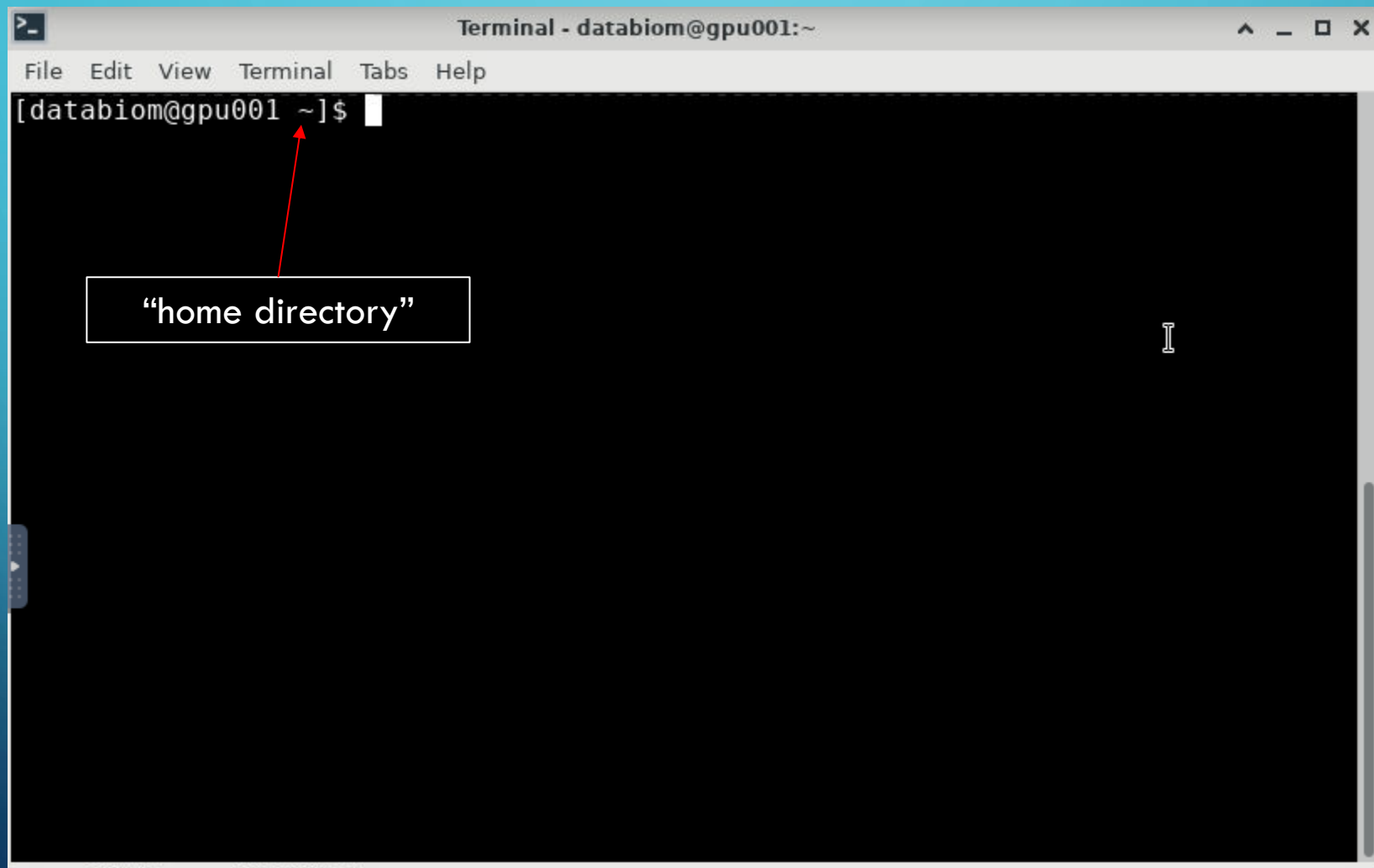
ALLOW COPY/PASTE



LINUX DESKTOP LAYOUT



TERMINAL



A terminal window titled "Terminal - databiom@gpu001:~" is shown. The window has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal content shows the prompt "[databiom@gpu001 ~]\$" followed by a cursor. A red arrow points from a text box labeled "home directory" to the tilde symbol (~) in the prompt, indicating that ~ represents the home directory.

```
Terminal - databiom@gpu001:~
File Edit View Terminal Tabs Help
[databiom@gpu001 ~]$
```

home directory

COMMON LINUX COMMANDS

- **ls** – “list” current working directory
- **cd** – “change directory”
 - **cd Directory1** → change/go to Directory1
 - **cd ..** → go back/up one directory
 - **cd** → will change/go to your home directory
- **mkdir** – “make directory”, create a new directory located in current directory
 - → **mkdir NewDirectory**

Everything is case sensitive!



COMMON LINUX COMMANDS

- `cp` → “copy” a file
- `mv` → “move” a file



These use the syntax
“*command file destination*”
e.g.: `cp file1 Directory1`

- `rm` → “remove” a file



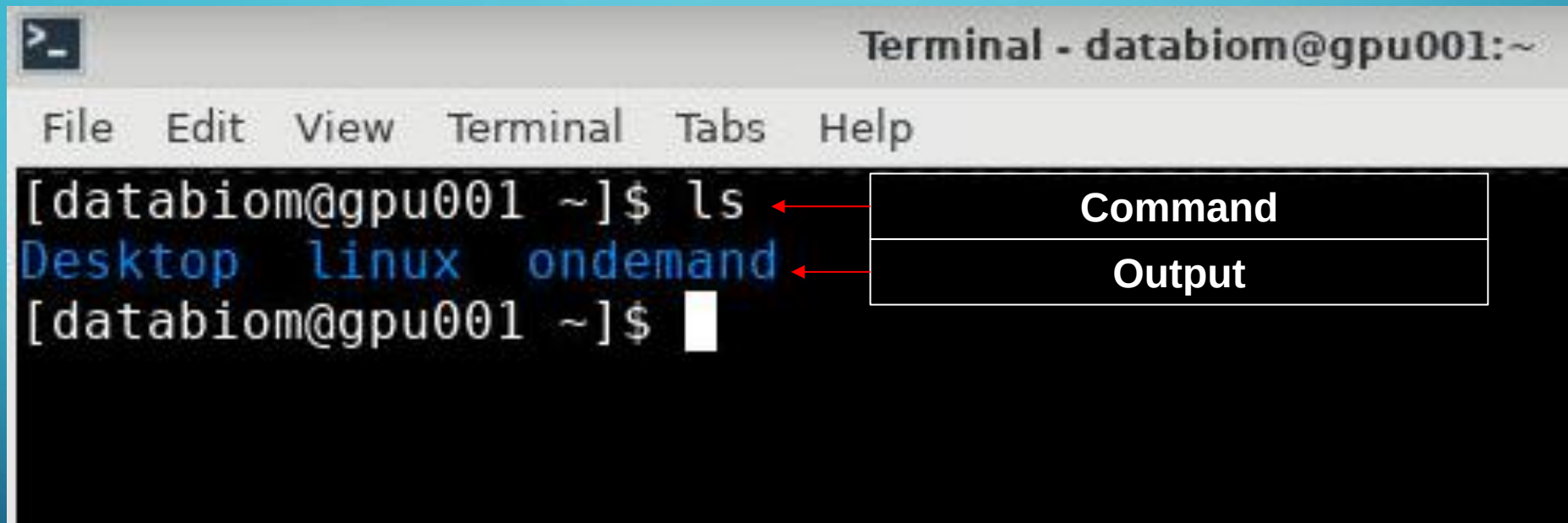
****There is no undo in linux****

- `head` → view first few lines of a file
- `tail` → view last few lines of a file
- `less` → view a file one page at a time (good for large files)



LIST CONTENTS OF CURRENT DIRECTORY

Everything is case sensitive!



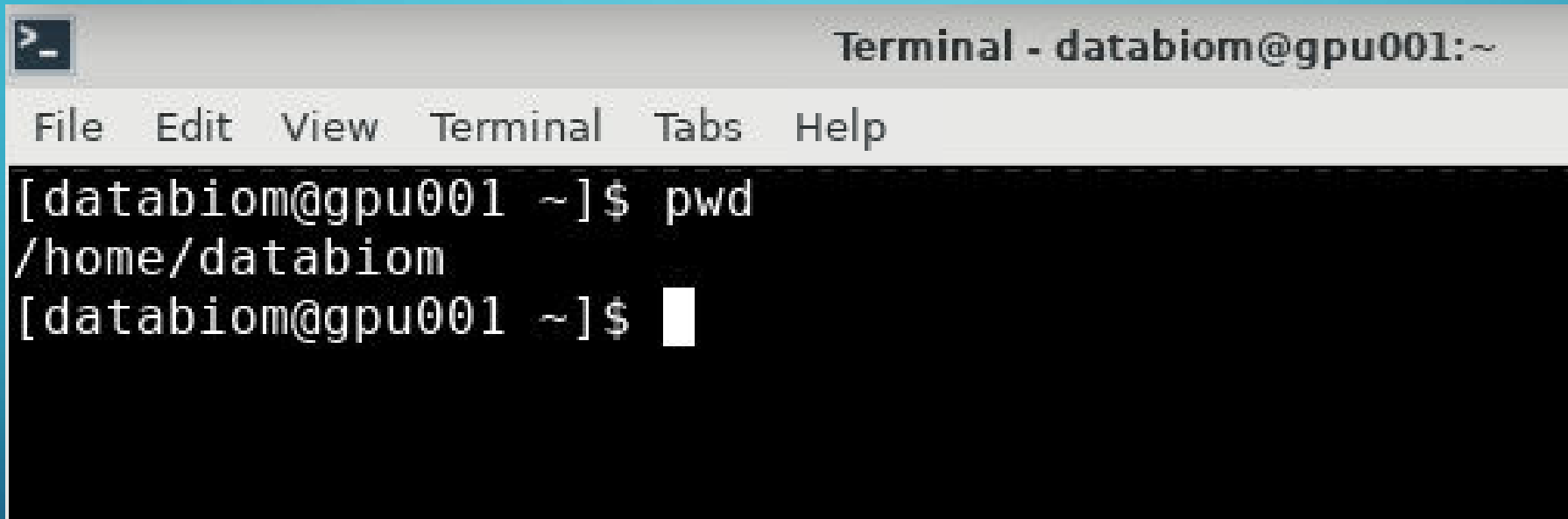
A terminal window titled "Terminal - databiom@gpu001:~" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The prompt is "[databiom@gpu001 ~]\$". The command "ls" has been entered, and the output "Desktop linux ondemand" is displayed on the next line. A white cursor is at the end of the prompt line. A table with two rows, "Command" and "Output", is overlaid on the right side of the terminal. Red arrows point from the "Command" row to the "ls" command and from the "Output" row to the output text.

Command
Output

*Hit 'Enter' to submit commands

GET PATH OF WORKING DIRECTORY

Everything is case sensitive!

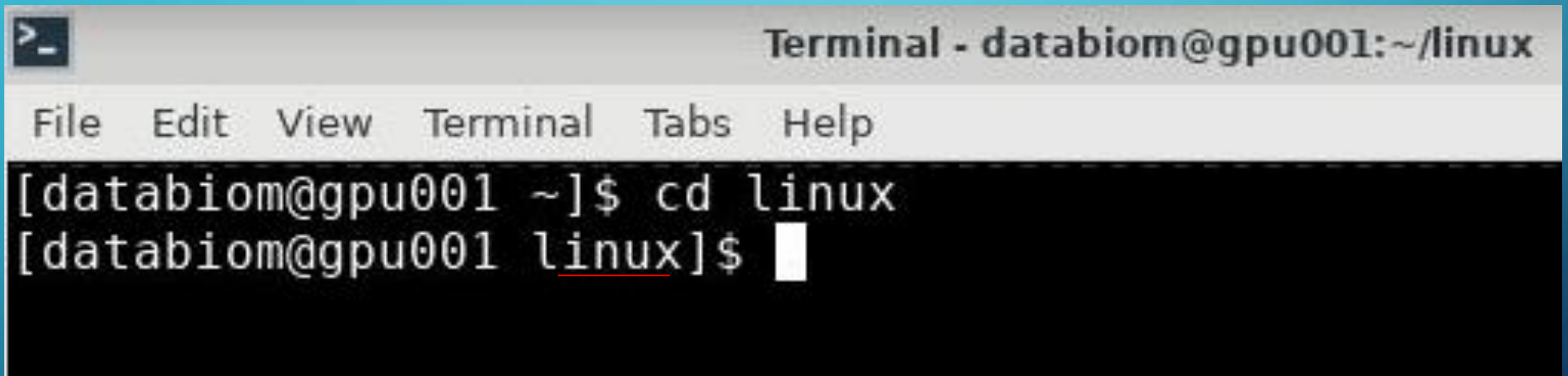
A terminal window titled "Terminal - databiom@gpu001:~" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows the command "[databiom@gpu001 ~]\$ pwd" and its output "/home/databiom". The prompt "[databiom@gpu001 ~]\$" is followed by a white cursor block.

```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 ~]$ pwd  
/home/databiom  
[databiom@gpu001 ~]$
```

MAKE A NEW DIRECTORY

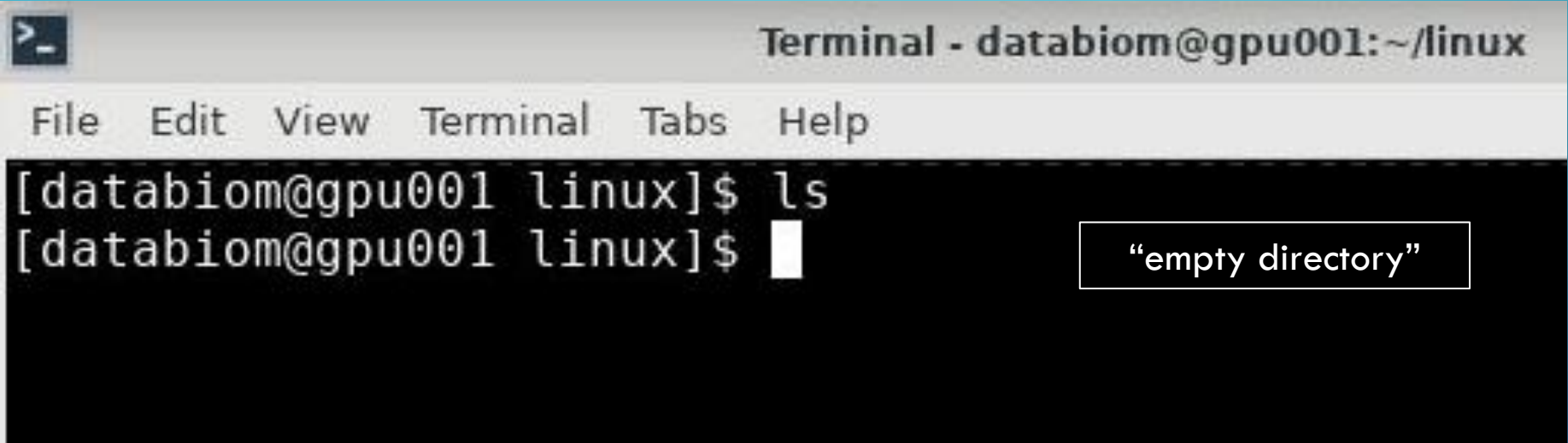
```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 ~]$ mkdir linux  
[databiom@gpu001 ~]$ ls  
Desktop linux ondemand  
[databiom@gpu001 ~]$
```

CHANGE (GO INTO A) DIRECTORY

A terminal window titled "Terminal - databiom@gpu001:~/linux" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows the command "cd linux" being executed, changing the directory from "~" to "linux". The prompt changes from "[databiom@gpu001 ~]" to "[databiom@gpu001 linux]", with "linux" underlined in red. A white cursor is visible after the second prompt.

```
>_ Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 ~]$ cd linux
[databiom@gpu001 linux]$
```

LIST CONTENTS (OR LACK OF) IN DIRECTORY

A terminal window titled "Terminal - databiom@gpu001:~/linux" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal shows the command "ls" being executed, resulting in no output, indicating an empty directory. A callout box points to the empty line after the command.

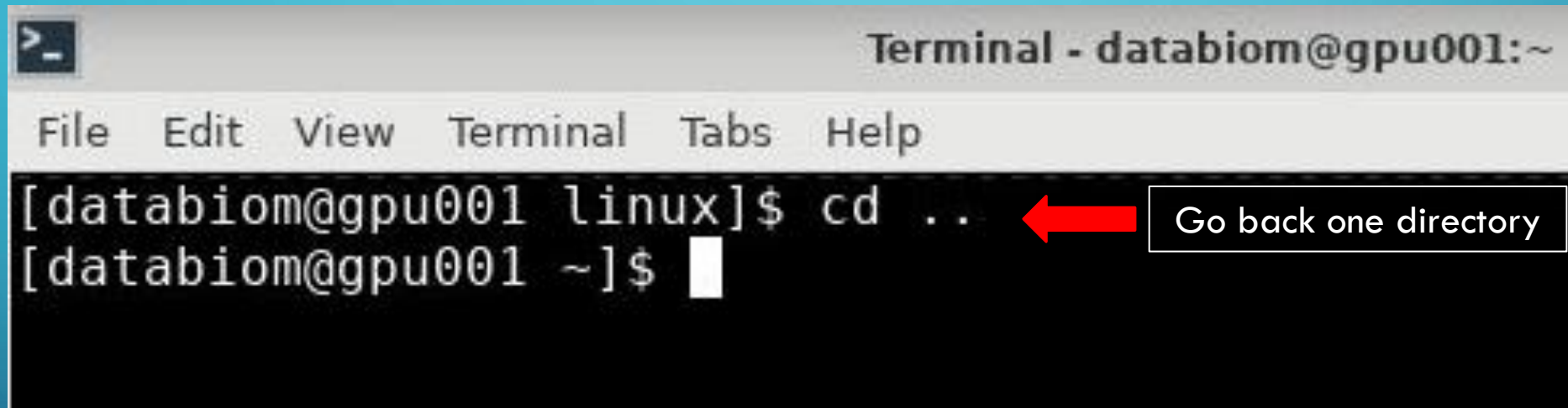
```
>_ Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ ls
[databiom@gpu001 linux]$
```

"empty directory"

DIRECTORY STRUCTURE/HIERARCHY

```
Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ pwd
/home/databiom/linux
[databiom@gpu001 linux]$
```

CHANGE DIRECTORY (GO BACK)

A terminal window titled "Terminal - databiom@gpu001:~" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows a command prompt where the user enters "cd ..". A red arrow points from a text box to the ".." part of the command. The prompt then changes to show the user is in the home directory (~).

```
>_ Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 linux]$ cd ..  
[databiom@gpu001 ~]$
```

Go back one directory

"cd" followed by nothing will take you back to your home directory

USING THE WGET COMMAND TO DOWNLOAD FROM INTERNET

```
Terminal - databiom@gpu011:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu011 ~]$ cd linux/
[databiom@gpu011 linux]$ wget https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/breast_cancer.txt
```

https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/breast_cancer.txt

DOWNLOAD INTO YOUR LINUX DIRECTORY

```
Terminal - databiom@gpu011:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu011 ~]$ cd linux/
[databiom@gpu011 linux]$ wget https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/breast_cancer.txt
--2025-01-26 17:10:52-- https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/breast_cancer.txt
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11366591 (11M) [text/plain]
Saving to: 'breast_cancer.txt.2'

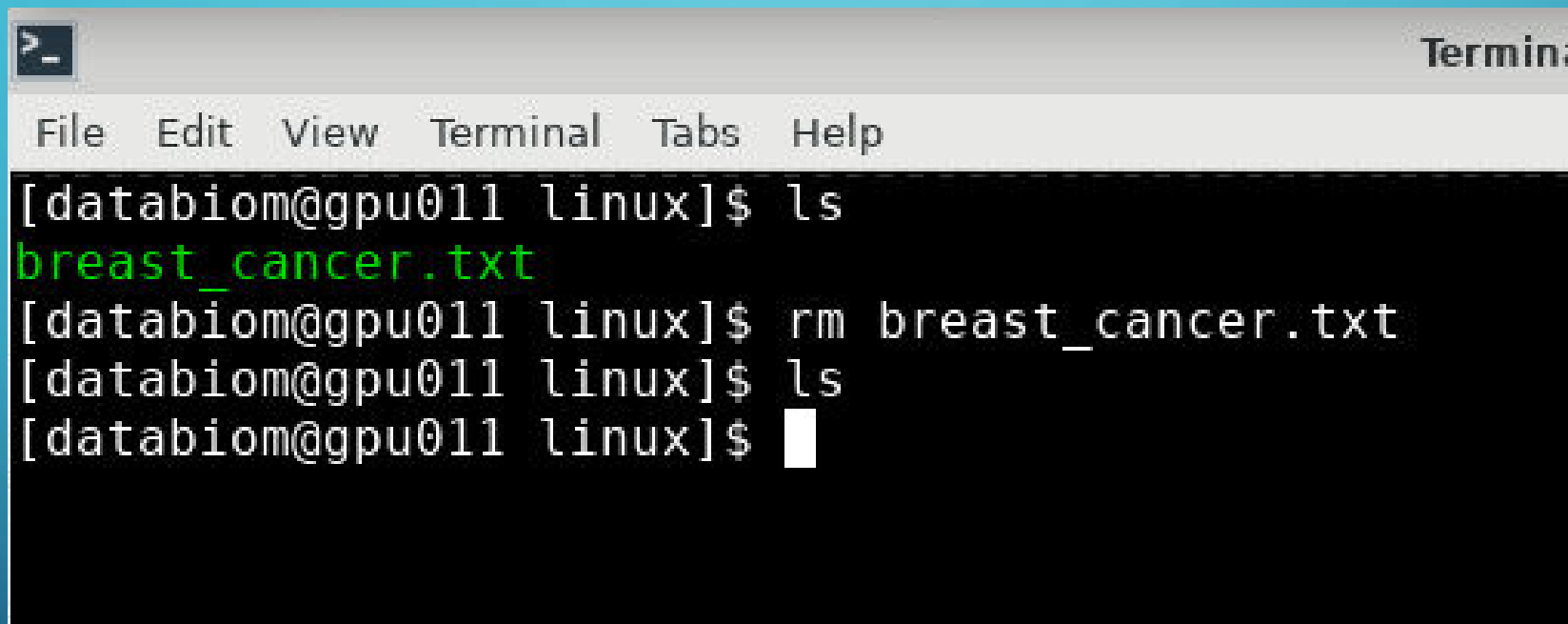
breast_cancer.txt.2      100%[=====>] 10.84M  67.7MB/s   in 0.2s

2025-01-26 17:10:52 (67.7 MB/s) - 'breast_cancer.txt.2' saved [11366591/11366591]

[databiom@gpu011 linux]$
```

“clear” will remove
everything above
if you want

REMOVING FILES

A terminal window titled 'Terminal' with a menu bar containing 'File', 'Edit', 'View', 'Terminal', 'Tabs', and 'Help'. The terminal shows a user named 'databiom' on a host named 'gpu011' in a 'linux' environment. The user runs 'ls' and sees 'breast_cancer.txt' in green. Then they run 'rm breast_cancer.txt' and 'ls' again, and the file is no longer listed.

```
[databiom@gpu011 linux]$ ls
breast_cancer.txt
[databiom@gpu011 linux]$ rm breast_cancer.txt
[databiom@gpu011 linux]$ ls
[databiom@gpu011 linux]$
```

COPYING FILES

```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 linux]$ cd ..  
[databiom@gpu001 ~]$ cp /home/databiom/linux/breast_cancer.txt .
```

Syntax: “*command [file] [destination]*”
e.g.: cp file1 Directory1



“copy to current
directory”

MOVING FILES

```
Terminal -
File Edit View Terminal Tabs Help
[databiom@gpu011 ~]$ ls
breast_cancer.txt Desktop linux ondemand
[databiom@gpu011 ~]$ mv breast_cancer.txt linux/
[databiom@gpu011 ~]$ cd linux/
[databiom@gpu011 linux]$ ls
breast_cancer.txt
[databiom@gpu011 linux]$
```

Syntax: “*command [file] [destination]*”
e.g.: `mv file1 Directory1`

USING A TEXT EDITOR TO READ AND EDIT FILES

```
>_
File Edit View Terminal Tabs Help
[databiom@gpu011 linux]$ nano breast_cancer.txt
```

Text editor

Text file

```
Terminal - databiom@gpu011:~/linux
GNU nano 2.9.8 breast_cancer.txt

Hugo_Symbol  Entrez_Gene_Id  Center  NCBI_Build  Chromosome  Start_Position  End_Position  Strand  Variant_Classifis
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
FOXA1 3169 MSKCC GRCh37 14 38061240 38061240 + Missense_Mutation SNP G G $
FOXA1 3169 MSKCC GRCh37 14 38061240 38061240 + Missense_Mutation SNP G G $
DICER1 23405 MSKCC GRCh37 14 95574698 95574698 + Missense_Mutation SNP C C $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
DICER1 23405 MSKCC GRCh37 14 95562249 95562249 + Missense_Mutation SNP C C $
DICER1 23405 MSKCC GRCh37 14 95562693 95562693 + Missense_Mutation SNP C C $
DICER1 23405 MSKCC GRCh37 14 95562249 95562249 + Missense_Mutation SNP C C $
DICER1 23405 MSKCC GRCh37 14 95562693 95562693 + Missense_Mutation SNP C C $
FOXA1 3169 MSKCC GRCh37 14 38060789 38060800 + In_Frame_Del DEL GAACGGGTGGTT GAACGGGT$
FOXA1 3169 MSKCC GRCh37 14 38061188 38061189 + In_Frame_Ins INS - - CTT $
AX 4149 MSKCC GRCh37 14 65544679 65544679 + Nonsense_Mutation SNP G G $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
NKX2-1 7080 MSKCC GRCh37 14 36987093 36987093 + Missense_Mutation SNP G G $
FOXA1 3169 MSKCC GRCh37 14 38060953 38060953 + Missense_Mutation SNP C C $
FOXA1 3169 MSKCC GRCh37 14 38060986 38060986 + Missense_Mutation SNP G G $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
AKT1 207 MSKCC GRCh37 14 105258970 105258970 + Missense_Mutation SNP A A $
FOXA1 3169 MSKCC GRCh37 14 38061231 38061231 + Missense_Mutation SNP A A $
MAX 4149 MSKCC GRCh37 14 65544703 65544703 + Nonsense_Mutation SNP G G $
MAX 4149 MSKCC GRCh37 14 65544703 65544703 + Nonsense_Mutation SNP G G $
AKT1 207 MSKCC GRCh37 14 105246436 105246436 + Missense_Mutation SNP A A $
FOXA1 3169 MSKCC GRCh37 14 38061007 38061022 + Frame_Shift_Del DEL GCCCGGGGGCCGGCGC $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
AKT1 207 MSKCC GRCh37 14 105239673 105239673 + Missense_Mutation SNP G G $
RAD51B 5890 MSKCC GRCh37 14 68331719 68331719 + Splice_Site SNP G G T $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $
AKT1 207 MSKCC GRCh37 14 105246551 105246551 + Missense_Mutation SNP C C $

[ Read 10196 lines (Converted from DOS format) ]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo M-A Mark Text
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line M-E Redo M-6 Copy Text
```

ctrl + x to exit

GREP COMMAND FOR FILTERING

“Pattern of interest”

“Filename”



A screenshot of a terminal window. The title bar at the top says "Terminal". Below the title bar is a menu bar with the options "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal prompt is "[databiom@gpu011 linux]\$". The command entered is "grep 'Y537S' breast_cancer.txt". Two red arrows point to the command: one points to the single quote character before 'Y537S', and the other points to the single quote character after 'Y537S'.

```
[databiom@gpu011 linux]$ grep "Y537S" breast_cancer.txt
```

“Pattern found”

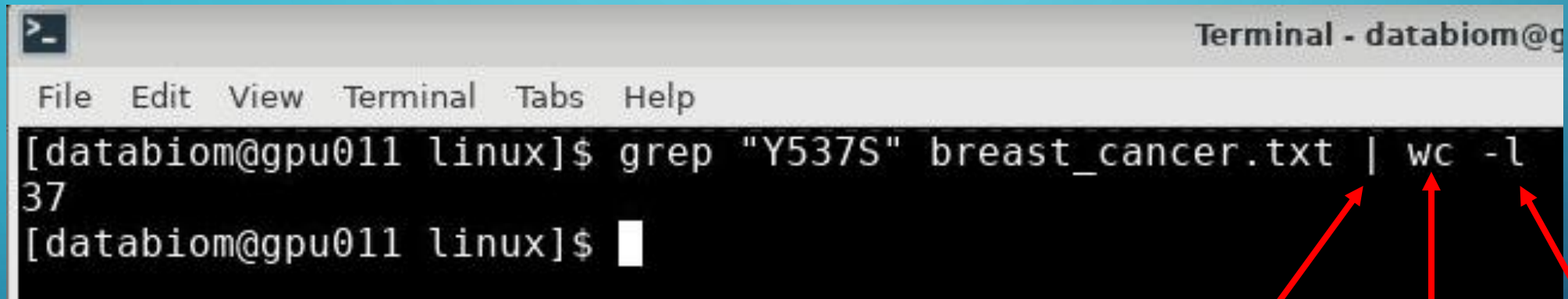
```

Terminal - databiom@gpu011:~/linux
File Edit View Terminal Tabs Help

A          "1,1"          NA      NA      NA      NA
ESR1      2099      MSKCC      GRCh37      6      152419923      152419923      +      Missense_Mutation      SNP      A      A      C
NA      P-0014995-T01-IM6      NORMAL      A      A      NA      NA      NA      NA      NA      Unknown SOMATIC      NA      NA      N
A      MSK-IMPACT      NA      NA      NA      NA      c.1610A>C      p.Tyr537Ser      p.Y537S      ENST00000440973      10/10      555      4
51      104      391      391      0      "ESR1,missense_variant,p.Tyr537Ser,ENST00000440973,NM_001122742.1;ESR1,missense_variant,p
.Tyr537Ser,ENST00000206249,NM_000125.3;ESR1,missense_variant,p.Tyr537Ser,ENST00000443427,NM_001122740.1;ESR1,missense_variant,p.T
yr425Ser,ENST00000456483,NM_001122741.1;ESR1,missense_variant,p.Tyr537Ser,ENST00000338799,;ESR1,missense_variant,p.Tyr276Ser,ENST
00000406599,;ESR1,intron_variant,ENST00000427531,;"      C      ENSG00000091831      ENST00000440973      Transcript      missense_variant1
980/6466      1610/1788      537/595      Y/S      tAt/tCt      "COSM1074637,COSM1074639"      1      NA      ESR1      HGNC      3467      p
rotein_coding      YES      CCDS5234.1      ENSP00000405330      P03372      "Q9UE35,Q9UBT1,Q5T5H8,K7R989,K4Q331,K4PXX0,K4PX46,G4XH65,E3WH18,C
8CJL0,C7FFR5,B6DU67"      UPI000004A328      NM_001122742.1      deleterious(0)      probably_damaging(0.996)      "Gene3D:1.10.565.
10,PIRSF_domain:PIRSF002527,PIRSF_domain:PIRSF500101,hmmpanther:PTHR24084,hmmpanther:PTHR24084:SF3,Superfamily_domains:SSF48508"N
A          "1,1"          NA      NA      NA      NA
ESR1      2099      MSKCC      GRCh37      6      152419923      152419923      +      Missense_Mutation      SNP      A      A      C
NA      P-0015097-T01-IM6      NORMAL      A      A      NA      NA      NA      NA      NA      Unknown SOMATIC      NA      NA      N
NA      MSK-IMPACT      NA      NA      NA      NA      c.1610A>C      p.Tyr537Ser      p.Y537S      ENST00000440973      10/10      646      5
96      50      391      391      0      "ESR1,missense_variant,p.Tyr537Ser,ENST00000440973,NM_001122742.1;ESR1,missense_variant,p
.Tyr537Ser,ENST00000206249,NM_000125.3;ESR1,missense_variant,p.Tyr537Ser,ENST00000443427,NM_001122740.1;ESR1,missense_variant,p.T

```

“PIPING” TO OTHER COMMANDS



A terminal window titled "Terminal - databiom@g" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The command `grep "Y537S" breast_cancer.txt | wc -l` is entered. The output `37` is displayed on the line below. The prompt `[databiom@gpu011 linux]$` is shown again on the next line.

```
[databiom@gpu011 linux]$ grep "Y537S" breast_cancer.txt | wc -l
37
[databiom@gpu011 linux]$
```

“Pipe”

“Word count”

“Count lines”

REDIRECT OUTPUT TO A NEW FILE



A terminal window titled "Terminal - databiom@gpu011:" shows the following commands and output:

```
[databiom@gpu011 linux]$ grep "Y537S" breast_cancer.txt > Y537S.txt
[databiom@gpu011 linux]$ nano Y537S.txt
```

Two red arrows point to the file names "Y537S.txt" in the command line.

[illegible]

“new file”

“redirect”

[illegible]

```
Terminal - databiom
File Edit View Terminal Tabs Help
[databiom@gpu011 linux]$ sed -i 's/ESR1/ERalpha/' Y537S.txt
```

“in place” “substitute” “old” “new”

[illegible]

EXTRACTING COLUMNS WITH AWK COMMAND

```
Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ awk '{print $9}' breast_cancer.txt > mutation_types.txt
[databiom@gpu001 linux]$ nano mutation_types.txt
```

“column number”

```
Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
GNU nano 2.9.8 mutation_types.txt
Variant_Classification
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
In_Frame_Del
In_Frame_Ins
onsense_Mutation
issense_Mutation
Missense_Mutation
Missense_Mutation
```

MODULES (PROGRAMS/APPLICATIONS)

Type 'module avail' to see all the available modules

```
Terminal - scott.widmann@gpu001:~
File Edit View Terminal Tabs Help
(base) [scott.widmann@gpu001 ~]$ module avail
----- /cm/local/modulefiles -----
boost/1.77.0          cmd          freeipmi/1.6.8    luajit          null          shared
cluster-tools-dell/9.2 cmjob        gcc/11.2.0        mariadb-libs   openldap      slurm/slurm/21.08.8
cluster-tools/9.2    cuda-dcgm/2.4.6.1 ipmitool/1.8.18  module-git     python3
cm-bios-tools        dot          lua/5.4.4         module-info    python39
----- /cm/shared/modulefiles -----
blacs/openmpi/gcc/64/1.1patch03 default-environment intel/compiler/64/2020/19.1.3
blas/gcc/64/3.10.0          fftw3/openmpi/gcc/64/3.3.10 intel/mpi/64/(default)
bonnie++/2.00a              gdb/10.2          intel/mpi/64/2020/4.304
cm-pmix3/3.1.4              globalarrays/openmpi/gcc/64/5.8 iozone/3_492
cm-pmix4/4.1.1              hdf5/1.12.1       jupyter-eg-kernel-wlm-py39/3.0.1
cuda11.0/blas/11.0.3         hdf5_18/1.8.21    jupyter/14.2.2
cuda11.0/fft/11.0.3          hwloc/1.11.11     lapack/gcc/64/3.10.0
cuda11.0/toolkit/11.0.3      hwloc2/2.7.1      mpich/ge/gcc/64/3.4.2
cuda11.8/blas/11.8.0         intel-cluster-runtime/ia32/2019.6 mvapich2/gcc/64/2.3.7
cuda11.8/fft/11.8.0          intel-cluster-runtime/ia32/2020.4(default) namd/20220721
cuda11.8/toolkit/11.8.0      intel-cluster-runtime/intel64/(default) netcdf/gcc/64/gcc/64/4.8.1
cuda12.0/blas/12.0.1         intel-cluster-runtime/intel64/2019.6 netperf/2.7.0
cuda12.0/fft/12.0.1          intel-cluster-runtime/intel64/2020.4(default) openblas/dynamic/(default)
cuda12.0/toolkit/12.0.1      intel-tbb-oss/ia32/2021.4.0 openblas/dynamic/0.3.18
cudnn8.6-cuda11.8/8.6.0.163 intel-tbb-oss/intel64/2021.4.0 openmpi/gcc/64/4.1.2
cudnn8.8-cuda12.0/8.8.1.3   intel/compiler/32/2020/19.1.3 openmpi4/gcc/4.1.2
cudnn8.9-cuda12.0/8.9.6.50  intel/compiler/64/(default) ucx/1.10.1
----- /apps/modulefiles -----
acpype/2022.7.21          cudnn/8.8.1.3      nextflow/23.04.02  rdkit/2023.3.2
adfrsuite/1.0             fastqc/0.12.1       nglview/3.0.6      ribodetector/0.2.7
amber/22                   gcc/12.3.0           openbabel/2.4.0    RSEM/1.3.3
anaconda/2023.03          git/2.41.0           openeye/2023.2.3   samtools/1.17
apptainer/1.1.9           grace/5.1.25         openmm/8.0.0        sas/9.4
autodock_gpu/1.5.1        gromacs/2023.1       parallel/20230722  scikit-learn/1.2.2
autodock_vina/1.1.2       gromacs_gpu/2023.1  perl/5.36.1         smog2/2.4.5
boost/1.82.0              gsea/4.3.2           ptmsi/py-3.9        sratoolkit/3.0.5
bowtie/2.5.1              gtk3/3.24.38         ptmsi/py-3.9-age    star/2.7.10b
bwa-mem2/2.2.1            htop/3.2.2           ptmsi_aqe/py-3.9    tensorflow-gpu/2.11.0-python-3.9.16
chimeraX/1                 lapack/3.11.0        python/3.11.2       torch/1.12.1-py3.9
chimeraX/1.6.1            libcusolver/11.4.4.55 QT/5.15.3           torch/2.0.1-py3.9
chimeraX/1.8              mgltools/1.5.7       qt/5.15.3           vmd/1.9.4
```

Need a particular program installed?
Request an installation.

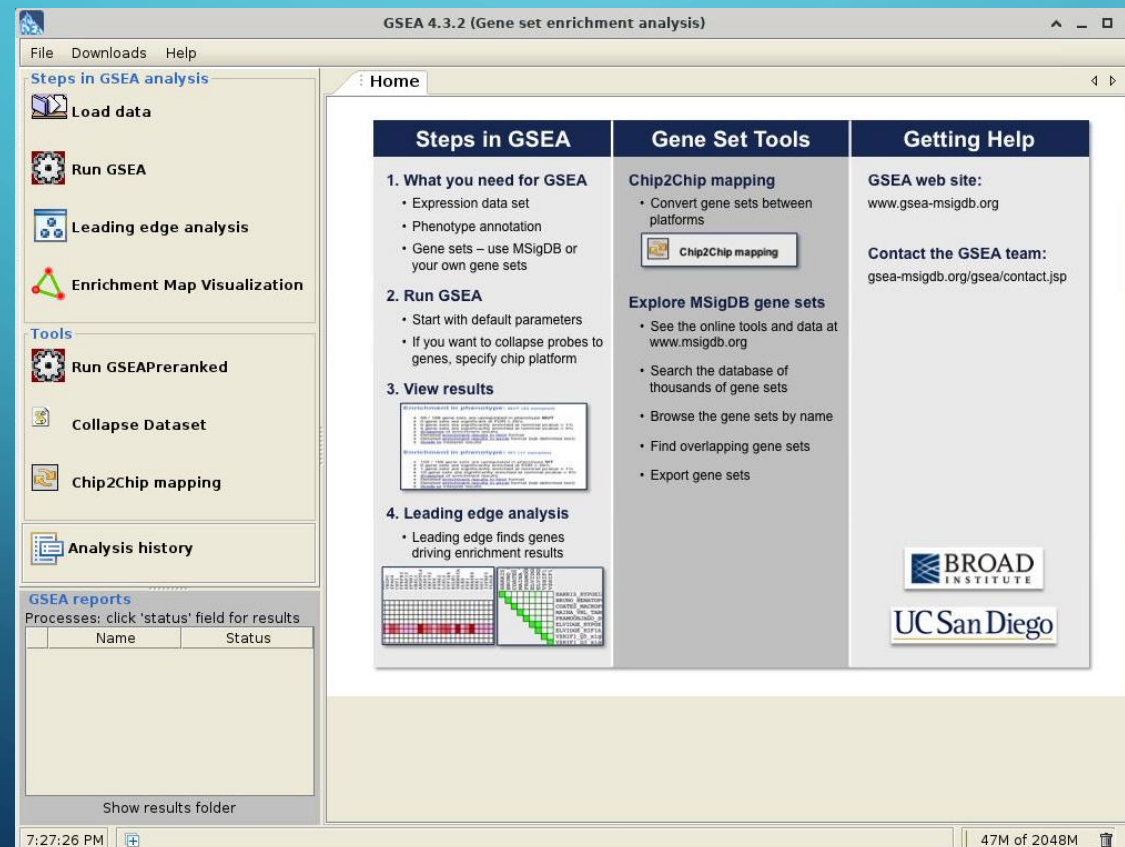
RUNNING A MODULE

```
Terminal -
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ module load gsea
[databiom@gpu001 linux]$ gsea.sh
```

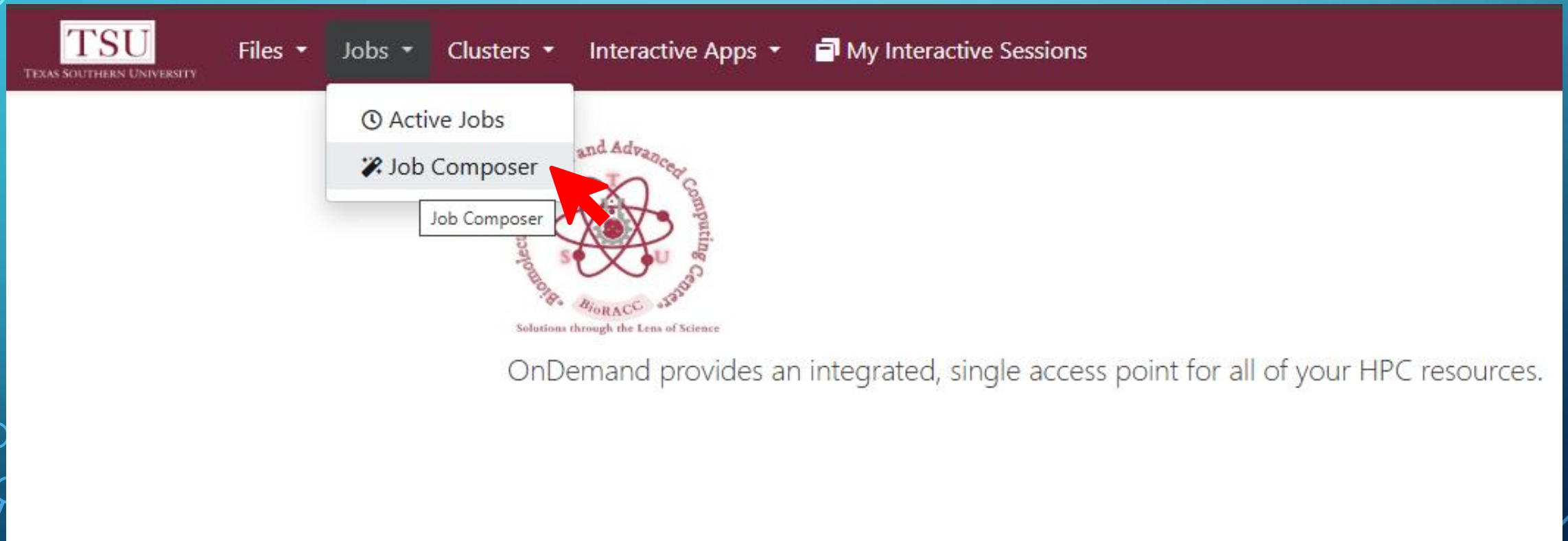
Must first load
module

This particular app
is started like this

Graphical User
Interface (GUI)
opens up



SUBMITTING A JOB



The screenshot displays the OnDemand web interface for Texas Southern University (TSU). The top navigation bar is dark red and contains the TSU logo, a 'Files' dropdown, a 'Jobs' dropdown, a 'Clusters' dropdown, an 'Interactive Apps' dropdown, and a 'My Interactive Sessions' link. The 'Jobs' dropdown menu is open, showing 'Active Jobs' and 'Job Composer' (highlighted with a red arrow). Below the menu, the BioRACC logo is visible, featuring an atom symbol and the text 'BioRACC' and 'Solutions through the Lens of Science'. The main content area contains the text: 'OnDemand provides an integrated, single access point for all of your HPC resources.'

TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ My Interactive Sessions

⌚ Active Jobs
✏ Job Composer
Job Composer

and Advanced Computing Center
BioRACC
Solutions through the Lens of Science

OnDemand provides an integrated, single access point for all of your HPC resources.

CREATE A NEW JOB

[Open OnDemand](#) / [Job Composer](#) [Jobs](#) [Templates](#)

Job was successfully destroyed.

Jobs

+ New Job ▾

From Default Template

From Specified Path

From Selected Job

☆ Create Template

Open Terminal

Submit

Stop

Delete

Search:

Created	Name	ID	Cluster	Status
No data available in table				
Showing 0 to 0 of 0 entries				

Previous

Next

NEW JOB

Open OnDemand / Job Composer Jobs Templates

Help

Job was successfully created.

Jobs

+ New Job

☆ Create Template

Edit Files

Job Options

Open Terminal

Submit

Stop

Delete

Show 25 entries

Search:

Created	Name	ID	Cluster	Status
January 26, 2025 7:33pm	(default) Simple Sequential Job		SLURM Cluster	Not Submitted

Showing 1 to 1 of 1 entries

Previous 1 Next

Job Details

Job Name:

(default) Simple Sequential Job

Submit to:

SLURM Cluster

Account:

Not specified

Script location:

/home/databiom/ondemand/data/sys/myjobs/projects/default/1

Script name:

main_job.sh

Folder Contents:

main_job.sh

JOB DETAILS

Job Details

Job Name:

(default) Simple Sequential Job

Submit to:

SLURM Cluster

Account:

Not specified

Script location:

/home/databiom/ondemand/data/sys/myjobs/projects/default/1

Script name:

main_job.sh

Folder Contents:

main_job.sh

Submit Script

main_job.sh

Script contents:

```
#!/bin/bash
# JOB HEADERS HERE

echo "Hello World"
```

Script that will be
executed

Prints "Hello World"

SUBMIT JOB

Job was successfully created.

Jobs

+ New Job ▾

☆ Create Template

📄 Edit Files

⚙️ Job Options

🔧 Open Terminal

Submit Job

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

Created



Name



ID



Cluster



Status



January 26, 2025 7:45pm

(default) Simple Sequential Job

SLURM Cluster

Not Submitted

Showing 1 to 1 of 1 entries

Previous

1

Next

JOB RUNNING

Jobs

+ New Job ▾

☆ Create Template

Edit Files

⚙ Job Options

🔗 Open Terminal

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

Created	⌵ Name	⌵ ID	⌵ Cluster	⌵ Status	⌵
January 26, 2025 7:45pm	(default) Simple Sequential Job	4215	SLURM Cluster	Running	

Showing 1 to 1 of 1 entries

Previous

1

Next

JOB COMPLETED

Jobs

+ New Job ▾

☆ Create Template

✎ Edit Files

⚙ Job Options

🔗 Open Terminal

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

Created	Name	ID	Cluster	Status
January 26, 2025 7:45pm	(default) Simple Sequential Job	4215	SLURM Cluster	Completed

Showing 1 to 1 of 1 entries

Previous 1 Next

CHECK OUTPUT OF JOB

Jobs

+ New Job ▾

☆ Create Template

Open Directory in File Manager

Edit Files

Job Options

Open Terminal

Submit

Stop

Delete

Show 25 ▾ entries

Search:

Created	Name	ID	Cluster	Status
January 26, 2025 7:45pm	(default) Simple Sequential Job	4215	SLURM Cluster	Completed

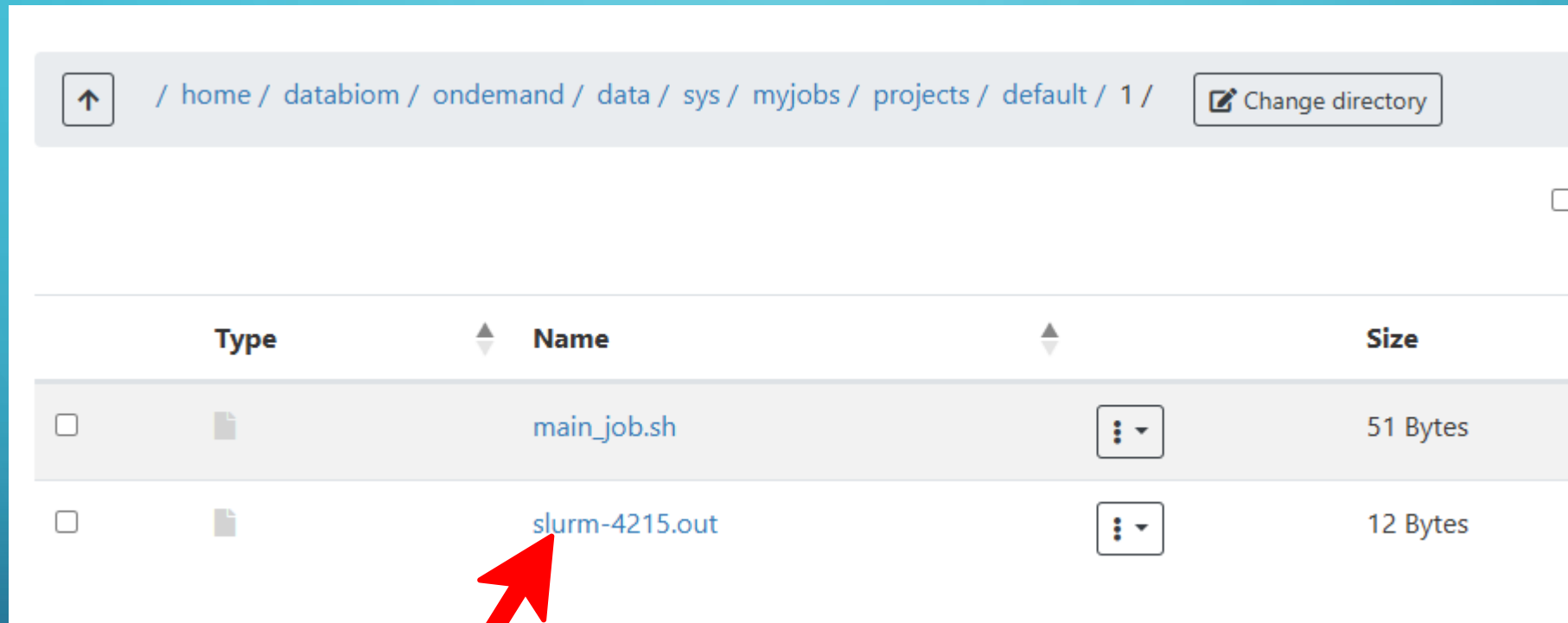
Showing 1 to 1 of 1 entries

Previous

1

Next

OUTPUT OF JOB



The screenshot shows a file manager interface with a breadcrumb path: / home / databiom / ondemand / data / sys / myjobs / projects / default / 1 /. There is a 'Change directory' button. Below the path is a table listing files:

Type	Name	Size
☐ 	main_job.sh	51 Bytes
☐ 	slurm-4215.out	12 Bytes

A red arrow points to the file 'slurm-4215.out'.

```
Hello World
```

PUBLISHED DATASET ON GEO

 **NCBI**


Gene Expression Omnibus

HOME | SEARCH | SITE MAP | GEO Publications | FAQ | MIAME | Email GEO

NCBI > GEO > **Accession Display** [?](#) Not logged in | [Login](#) [?](#)

Scope: Format: Amount: GEO accession:

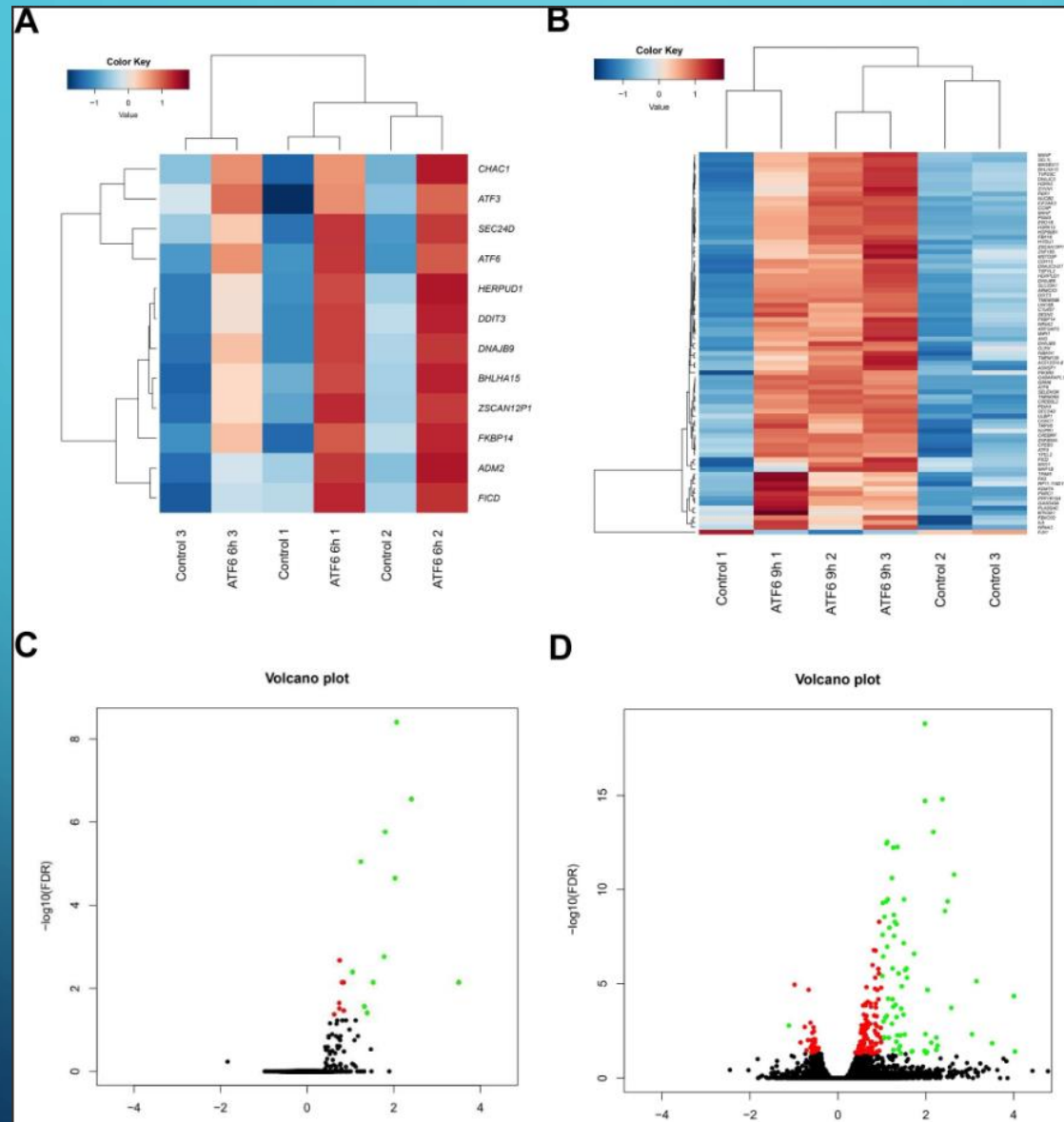
Series **GSE270286**

[Query DataSets for GSE270286](#)

Status	Public on Nov 06, 2024
Title	Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing
Summary	To investigate the molecular mechanism that triggers endoplasmic reticulum (ER) stress-induced cellular senescence in breast cancer cells, we established MCF7 cell lines exposed to ATF6a ectopic expression. We then performed gene expression profiling analysis using data obtained from RNA-seq of 9 different cells at three time points.
Overall design	Comparative gene expression profiling analysis of RNA-seq data for MCF-7 cells from three distinct time points (baseline, 6 hours, and 9 hours after ATF6a ectopic expression), using Illumina HiSeq X10 and NovaSeq 6000 platforms with paired-end 100 base sequencing.
Contributor(s)	Kim J , Bae S , Moon Y , Kim E , Kim Y , Park Y , Han M , Sohn J
Citation(s)	Kim JW, Bae SH, Moon Y, Kim EK et al. Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells. <i>PLoS One</i> 2024;19(10):e0309749. PMID: 39466820

Series accession
number

LET'S TRY TO REPRODUCE THEIR RESULTS



SEQUENCE READ ARCHIVE (SRA) RUN SELECTOR TO SEE SEQUENCING FILES

Supplementary file	Size	Download	File type/resource
GSE270286_RAW.tar	48.1 Mb	(http) (custom)	TAR (of TXT)

[SRA Run Selector](#) 

Raw data are available in SRA

ACCESSION IDS

Found 9 Items										
☒ ☐	Run	BioSample	Bases	Bytes	Experiment	Instrument	Library Name	create_date	Sample Name	treatment
☐ 1	SRR29476514	SAMN41934402	6.33 G	1.84 Gb	SRX24988435	Illumina NovaSeq 6000	GSM8338904	2024-06-20 14:05:00Z	GSM8338904	ATF6alpha overexpression for 9 hours
☐ 2	SRR29476515	SAMN41934403	6.32 G	1.83 Gb	SRX24988434	HiSeq X Ten	GSM8338903	2024-06-20 14:07:00Z	GSM8338903	ATF6alpha overexpression for 9 hours
☐ 3	SRR29476516	SAMN41934404	5.44 G	1.58 Gb	SRX24988433	HiSeq X Ten	GSM8338902	2024-06-20 14:05:00Z	GSM8338902	ATF6alpha overexpression for 9 hours
☐ 4	SRR29476517	SAMN41934405	6.32 G	1.85 Gb	SRX24988432	Illumina NovaSeq 6000	GSM8338901	2024-06-20 14:05:00Z	GSM8338901	ATF6alpha overexpression for 6 hours
☐ 5	SRR29476518	SAMN41934406	5.95 G	1.74 Gb	SRX24988431	HiSeq X Ten	GSM8338900	2024-06-20 14:04:00Z	GSM8338900	ATF6alpha overexpression for 6 hours
☐ 6	SRR29476519	SAMN41934407	4.94 G	1.46 Gb	SRX24988430	HiSeq X Ten	GSM8338899	2024-06-20 14:04:00Z	GSM8338899	ATF6alpha overexpression for 6 hours
☐ 7	SRR29476520	SAMN41934408	6.37 G	1.83 Gb	SRX24988429	Illumina NovaSeq 6000	GSM8338898	2024-06-20 14:07:00Z	GSM8338898	non-ATF6alpha overexpression
☐ 8	SRR29476521	SAMN41934409	6.09 G	1.78 Gb	SRX24988428	HiSeq X Ten	GSM8338897	2024-06-20 14:06:00Z	GSM8338897	non-ATF6alpha overexpression
☐ 9	SRR29476522	SAMN41934410	4.82 G	1.41 Gb	SRX24988427	HiSeq X Ten	GSM8338896	2024-06-20 14:06:00Z	GSM8338896	non-ATF6alpha overexpression

Individual
sequencing runs

Important
information
regarding samples

SRA TOOLKIT



[Systems](#) [Applications](#) [Reference Data](#) [Storage](#) [User Guides](#) [Training](#) [User Dashboard](#) [How To](#) [About](#)

SRA-Toolkit

The [NCBI SRA SDK](#) generates loading and dumping tools with their respective libraries for building new and accessing existing runs.

Documentation

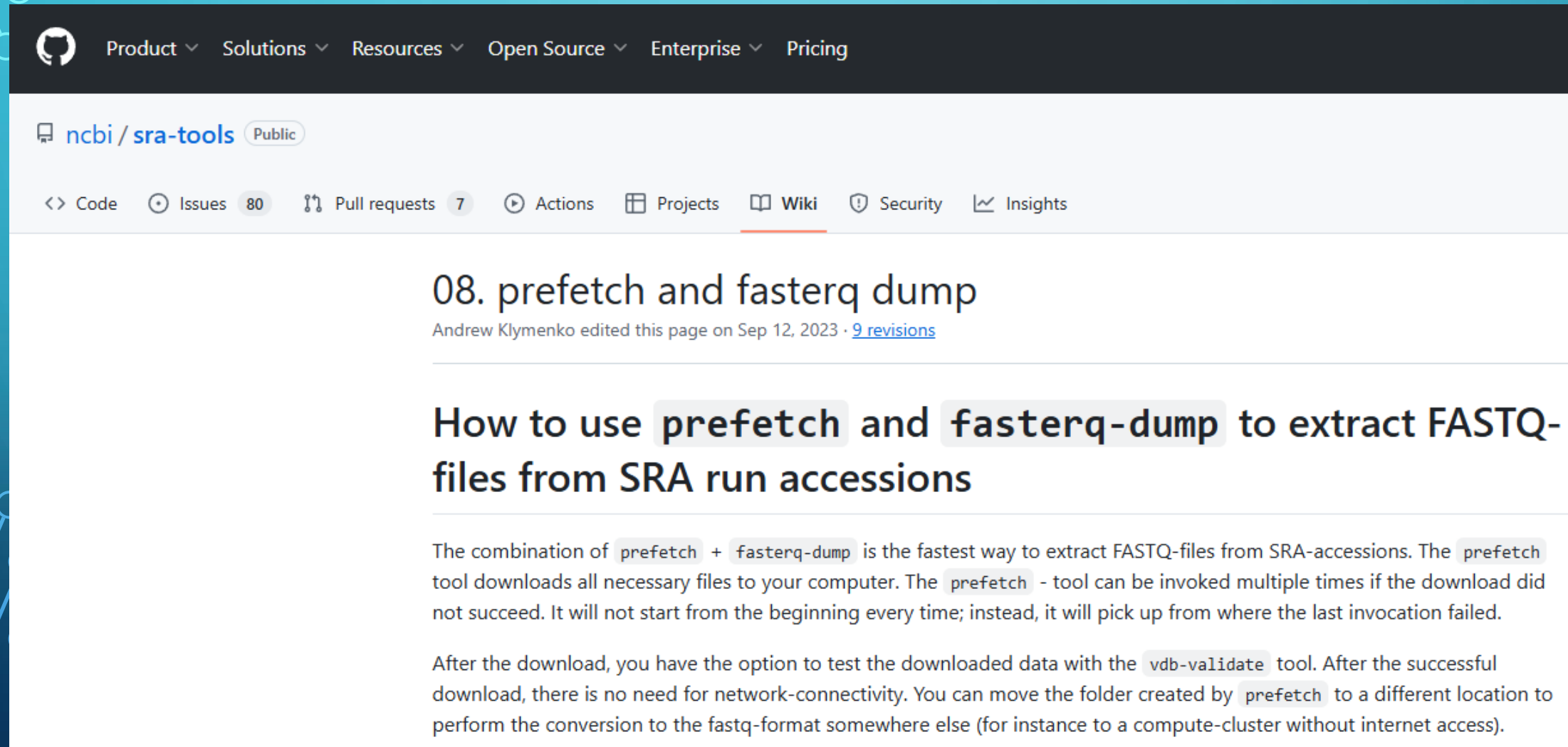
- [NCBI SRA Download Guide](#)
- [SRA Toolkit documentation](#)
- [SRA File Formats Guide](#)
- **Command line help:** Type the command followed by '-h'
- [fasterq-dump guide](#)

Quick Links

[SRA Source Repositories](#)
[Estimating space requirements](#)
[Downloading from NCBI](#)
[Batch job on Biowulf](#)
[Using Swarm](#)
[Interactive jobs](#)
[dbGaP downloads](#)
[Executables](#)
[Documentation](#)
[Configuring SRA-Toolkit](#)
[Troubleshooting](#)

Important Notes

PREFETCH AND FASTERQ-DUMP



The screenshot shows the GitHub interface for the repository `ncbi/sra-tools`. The top navigation bar includes links for Product, Solutions, Resources, Open Source, Enterprise, and Pricing. Below this, the repository name `ncbi/sra-tools` is displayed with a 'Public' label. A secondary navigation bar contains icons and labels for Code, Issues (80), Pull requests (7), Actions, Projects, Wiki (which is the active tab), Security, and Insights.

08. prefetch and fasterq dump

Andrew Klymenko edited this page on Sep 12, 2023 · [9 revisions](#)

How to use `prefetch` and `fasterq-dump` to extract FASTQ-files from SRA run accessions

The combination of `prefetch` + `fasterq-dump` is the fastest way to extract FASTQ-files from SRA-accessions. The `prefetch` tool downloads all necessary files to your computer. The `prefetch` - tool can be invoked multiple times if the download did not succeed. It will not start from the beginning every time; instead, it will pick up from where the last invocation failed.

After the download, you have the option to test the downloaded data with the `vdb-validate` tool. After the successful download, there is no need for network-connectivity. You can move the folder created by `prefetch` to a different location to perform the conversion to the fastq-format somewhere else (for instance to a compute-cluster without internet access).

USE A SCRIPT TO DOWNLOAD SEQUENCING DATA

Host: localhost

GNU nano 2.9.8

```
#!/bin/bash
```

Specifies 'bash' language of script

```
#SBATCH -J GSE270286
```

```
#SBATCH -o GSE270286_sra.out
```

```
#SBATCH -N 1 -n 32
```

- Job name
- Log file produced
- Requesting one whole node (32 cores)

```
module load sratoolkit
```

Loads the required sratoolkit module

```
while read p; do
```

```
    prefetch "$p"
```

```
    fasterq-dump -e 32 --outdir ./"$p"/ "$p"
```

```
    pigz -p 32 ./"$p"/*.fastq
```

```
    rm ./"$p"/*.sra
```

```
done < GSE270286.txt
```

USE A SCRIPT TO DOWNLOAD SEQUENCING DATA

Host: localhost

GNU nano 2.9.8

```
#!/bin/bash
```

```
#SBATCH -J GSE270286
```

```
#SBATCH -o GSE270286_sra.out
```

```
#SBATCH -N 1 -n 32
```

```
module load sratoolkit
```

```
while read p; do
```

```
    prefetch "$p"
```

```
    fasterq-dump -e 32 --outdir ./"$p"/ "$p"
```

```
    pigz -p 32 ./"$p"/*.fastq
```

```
    rm ./"$p"/*.sra
```

```
done < GSE270286.txt
```

'while loop' that reads each line of GSE270286.txt and replaces the variable 'p' with the accession run (SRR...)

Input for 'while loop'

- Prefetch each accession run ("p" variable)
- download each run (in parallel (32 cores))
- Unzip each run (in parallel)
- Remove unnecessary files

COPY OVER SCRIPT AND SRR LIST TO YOUR DIRECTORY

Script:

```
[databiom@tsu-hpc01 linux]$ cp /home/databiom/linux/sratoolkit_prefetch_dump_batch.sh .
```

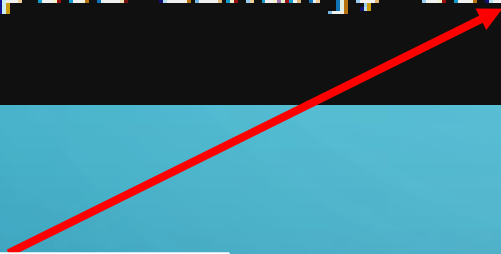
SRR List:

```
[databiom@tsu-hpc01 linux]$ cp /home/databiom/linux/GSE270286.txt .
```

SUBMIT JOB TO EXECUTE SCRIPT

```
[databiom@tsu-hpc01 linux]$ sbatch sratoolkit_prefetch_dump_batch.sh
```

Command to
submit job to job
scheduler



USE THE 'QUEUE' COMMAND TO CHECK THE JOB QUEUE FOR YOUR JOB

```
[databion@tsu-hpc01 linux]$ squeue
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	ODELIST(Reason)
4220	defq	gpu_md	kehinde.	R	3:13:04	1	gpu003
4221	defq	gpu_md	kehinde.	R	3:12:23	1	gpu004
4222	defq	gpu_md	kehinde.	R	3:11:41	1	gpu005
4223	defq	gpu_md	kehinde.	R	3:10:50	1	gpu006
4224	defq	gpu_md	kehinde.	R	3:10:14	1	gpu007
4225	defq	gpu_md	kehinde.	R	3:09:23	1	gpu008
4226	defq	gpu_md	kehinde.	R	3:08:24	1	gpu009
4227	defq	gpu_md	kehinde.	R	3:07:26	1	gpu010
4229	defq	::app=Li	kehinde.	R	50:56	1	gpu001
4230	defq	::app=Li	ceci	R	48:48	1	gpu011
4250	defq	gpu_md	kehinde.	R	15:02	1	gpu012
4251	defq	gpu_md	kehinde.	R	14:56	1	gpu013
4252	defq	gpu_md	kehinde.	R	14:50	1	gpu014
4253	defq	gpu_md	kehinde.	R	14:46	1	gpu015

```
[databion@tsu-hpc01 linux]$
```



GOOD JOB!

LINUX & HPC